Product Experimentation Platform

Overview

Product Experimentation Platform is a React based product analytics dashboard designed to simulate and evaluate A/B experiments. The project helps product managers compare two product variants, understand conversion performance, estimate business impact, evaluate statistical significance, monitor guardrail metrics, and make a product rollout decision.

I built this project to demonstrate how product experimentation can connect product metrics with business outcomes and practical product decision making.

Problem Statement

Product teams often need to decide whether a new product experience should be rolled out to users. Looking only at conversion rate is not enough because a change can improve conversion while creating negative effects elsewhere in the customer journey.

This project provides a single dashboard for evaluating an experiment across conversion, revenue, funnel performance, statistical significance, returns, cancellations, and overall product impact.

Project Objective

The main objective of the project is to build a practical experimentation platform that helps answer the following questions.

Which variant performs better

How much does conversion change between the variants

What is the estimated revenue impact

Where does the user funnel improve or decline

Is the observed difference statistically significant

Are there negative effects on important guardrail metrics

What should a product manager do next

Key Features

Experiment Setup: Users can configure the experiment by entering the total number of users, expected conversion rates, average order value, return rates, and cancellation rates for both variants.

A/B Experiment Simulation: The application randomly simulates user behavior and generates observed results for Variant A and Variant B.

Conversion Analysis: The dashboard calculates the observed conversion rate for each variant and calculates the relative conversion uplift between the two variants.

Revenue Impact: The project estimates revenue for both variants and calculates additional orders, additional revenue, and revenue uplift associated with Variant B.

Conversion Funnel: The experiment includes Visitors, Product Views, Add to Bag, Checkout, and Purchase stages. The dashboard calculates conversion rates between each stage.

Statistical Significance: The project uses a two proportion z test to compare observed conversion rates. A p value is calculated and compared against a significance threshold of 0.05. The dashboard also calculates 95 percent

confidence intervals.

Guardrail Metrics: The platform tracks return rate and cancellation rate to identify situations where a variant improves conversion but creates negative downstream effects.

Product Manager Decision Engine: The decision engine evaluates statistical significance, conversion performance, return rate, and cancellation rate. Possible recommendations are Roll Out Variant B, Investigate Variant B, Continue Experiment, and Do Not Roll Out.

PM Insights: The dashboard converts experiment results into product focused insights covering conversion performance, revenue impact, guardrail changes, recommended next steps, and overall product implications.

Experiment History: Every experiment run is stored in the current application session and displayed with date, winner, conversion rates, uplift, p value, and PM decision.

Interactive Data Visualization: Interactive charts visualize conversion comparison, funnel performance, and revenue comparison.

Technology Stack

```
React

Vite

JavaScript

CSS

Recharts
```

Project Structure

```
frontend

src

App.jsx

App.css

main.jsx

public

package.json
```

Getting Started

Clone the repository and open the project directory.

Install the required dependencies.

```
npm install
```

Start the development server.

```
npm run dev
```

Open the local development URL shown in the terminal.

How the Experiment Works

The application divides the configured users between Variant A and Variant B.

User behavior is simulated through the conversion funnel.

Product views are simulated first, followed by add to bag actions, checkout actions, and purchases.

The resulting purchase counts are used to calculate observed conversion rates.

The application then calculates revenue impact, statistical significance, confidence intervals, and guardrail differences.

Finally, the Product Manager Decision Engine combines the results and generates a recommended action.

Product Decision Framework

If the result is not statistically significant, the recommendation is to continue the experiment and collect more data.

If Variant B improves conversion and does not worsen the defined guardrail metrics, the recommendation is to roll out Variant B.

If Variant B improves conversion but worsens one or more guardrail metrics, the recommendation is to investigate Variant B before rollout.

If Variant B does not provide a meaningful improvement, the recommendation is not to roll it out.

Design

The interface follows a futuristic product analytics design with a dark visual theme, glass style panels, gradient accents, interactive states, and responsive layouts.

The visual design is intended to make a data heavy experimentation workflow easier to understand while maintaining a professional product dashboard experience.

Product Management Concepts Demonstrated

A/B experimentation

Hypothesis driven product decisions

Conversion rate analysis

Relative uplift

Revenue impact analysis

Conversion funnel analysis

Statistical significance

Confidence intervals

Guardrail metrics

Experiment decision making

Product insights

Follow up experimentation

Business impact analysis

Future Improvements

Possible future improvements include user segmentation, experiment configuration persistence, authentication, database storage, real analytics data integration, experiment sample size estimation, automated experiment reports, and deployment with a backend service.

Author

Built as a Product Management and software development portfolio project.

The project demonstrates my approach to combining product thinking, experimentation, data analysis, business impact measurement, and frontend development in one practical application.